

ATTACHMENT (1)
TECHNICAL ORDERING DATA (TOD)

CONTRACT No: N3904025P0138
ITEM: 0001

SECTION B - SUPPLIES/SERVICES:

NATIONAL STOCK NUMBER (NSN): 1H 4710-00-455-0193 X2

ITEM NAME: TUBE, COPPER-NICKEL, COMPOSITION 70-30, SEAMLESS, ANNEALED

PART NUMBER: MILT16420K,0.375OD

QUANTITY: As specified in the contract/delivery order.

SPECIAL CHARACTERISTICS: CLASS 3300, O.D. 0.375-INCH, MIN. WALL THICKNESS 0.049-INCH

SECTION C - DESCRIPTION/SPECIFICATIONS:

C.1. Manufacture per the attached Individual Repair Part Ordering Data (IRPOD), as modified by this contract.

C.1.1. The order of precedence for any conflicting documentation/requirements shall be as follows:

- (a) The contract (including attachments, modifications, amendments, etc.).
- (b) The IRPOD.
- (c) Drawings (if any) referenced by the IRPOD.
- (d) Specifications referenced by the IRPOD or drawing.

C.1.2. Manufacture in accordance with MIL-T-16420, except as modified herein. Ordering Data for paragraph 6.1 of MIL-T-16420 is as follows:

- (a) See section C.2.1 for specification information.
- (b) Composition: 70/30 (Copper Alloy No. 715), Temper: Annealed, Type I (Seamless), Grade 2 (except that each lot shall contain one heat), Class to be as specified in Section B of this Schedule.
- (c) Hydrostatic test is required for tube that is ultrasonically tested.
- (d) Not applicable.
- (e) Not applicable.
- (f) Not applicable.
- (g) Not applicable.
- (h) See Section B of this schedule.
- (i) Dimensions shall be specified in Section B of this schedule. Weight as specified in Table III of MIL-T-16420 with the following modifications:
 - (1) 0.250" Nominal O.D. Tube, NSN 1H 4710-00-455-0192 X2 shall be special weight as follows:
Pounds/foot (calculated) = 0.119
Pounds/foot (max) = 0.132
 - (2) 0.500" Nominal O.D. Tube, NSN 1H 4710-00-432-3098 X2 shall be special weight as follows:
Pounds/foot (calculated) = 0.271
Pounds/foot (max) = 0.298
 - (3) 1.315" Nominal O.D. Tube, NSN 1H 4710-00-432-3125 X2 shall be special weight as follows:
Pounds/foot (calculated) = 1.76
Pounds/foot (max) = 1.93
 - (4) 3.500" Nominal O.D. Tube, LSN 4710-LL-RT1-3000 shall be special weight as follows:
Pounds/foot (calculated) = 11.691
Pounds/foot (max) = 12.860

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- (j) Mill lengths in accordance with paragraph 3.8.5.2 of MIL-T-16420, except maximum lengths of 20 feet are preferable.
- (k) Marking per paragraph 3.10 of MIL-T-16420 and as specified in C.8.
- (l) Paragraph 4.1.1 of MIL-T-16420 applies.
- (m) Hydrostatic test pressure for class special is as follows:

Size	Pressure (min/max)
1.315 x 0.120	1950/2100
0.500 x 0.049	3740/4110
0.250 x 0.049	8010/8807
3.500 x 0.300	3254/3579

(n) See section D of this schedule.

(o) Not required.

C.1.3. Modify MIL-T-16420K as follows:

- (a) Delete para 3.4.3.4 and substitute:

“3.4.3.4 Laminations. Tube required to meet the ultrasonic requirements specified in 3.4.3 shall be examined for laminar discontinuities by ultrasonic gageing. Tube containing laminar discontinuities that present indications exceeding exploration threshold levels (20 percent for the contact method at +6db, 30 percent for the immersion method at calibration) shall be rejected.”

- (b) In the second sentence of paragraph 4.4.7, delete “18 inch” and substitute “1 inch”. In the last sentence, insert “3.4.3.4 and” before “3.8.2”.

C.1.4. NAVSEA Technical Publication T9074-AS-GIB-010/271 supersedes MIL-STD-271, which has been cancelled. All references to MIL-STD-271 shall be replaced with T9074-AS-GIB-010/271.

- (a) T9074-AS-GIB-010/271 REV 1 of 9/11/2014 shall be used.

C.1.5. Each finished tube shall be 100% ultrasonically examined from the outside diameter per T9074-AS-GIB-010/271 to assure compliance with the wall thickness requirements of Table III or IV of MIL-T-16420K.

- (a) The vendor shall notify the purchasing agent four (4) weeks in advance of ultrasonic testing to allow participation of Portsmouth Naval Shipyard personnel in witnessing the ultrasonic tests.

C.1.6. Supplemental Technical Requirements (STR) 505 Revision 3 is invoked. STR-505 can be obtained from Portsmouth Naval Shipyard upon request.

C.2. APPLICABLE DOCUMENTS:

C.2.1. Section 2.0 of the IRPOD lists applicable documents to be used (e.g., drawings, specifications, standards, etc.) and their required version level (e.g., revisions, amendments, change notices, interim changes, etc.). The following list of documents and version information takes precedence over version information (including the lack of any version information) identified by the IRPOD:

- (a) MIL-T-16420K with AMENDMENT 1; TUBE, COPPER-NICKEL ALLOY, SEAMLESS AND WELDED)
(COPPER ALLOY NUMBERS 715 and 706)

- (b) MIL-STD-792F with CHANGE 1; IDENTIFICATION MARKING REQUIREMENTS FOR SPECIAL PURPOSE COMPONENTS
- (c) MIL-STD-129R, CHANGE 2; MARKING FOR SHIPMENT AND STORAGE
- (d) ASTM D3951-18; STANDARD PRACTICE FOR COMMERCIAL PACKAGING
- (e) NAVSEA TECHNICAL PUBLICATION T9074-AS-GIB-010/271 REV 1; REQUIREMENTS FOR NONDESTRUCTIVE TESTING METHODS

C.2.2. If no document version is identified (anywhere in the contract or referenced documents), then use the latest version identified in the Acquisition Streamlining and Standardization Information System (ASSIST) at the time of contract award; ASSIST information is available at the following website:

<https://quicksearch.dla.mil>. If no document version information is available in ASSIST, then any version of the document may be used provided its version is dated 1 November 1969 or later.

C.2.3. In any case, the vendor shall use only one version in its entirety (i.e., the vendor shall not use portions of different versions of a document).

C.3. DATA REQUIREMENTS:

C.3.1. Material Certification; See Exhibit A, Data Item No. A001.

C.3.2. Procedure Approvals; See Exhibit A, Data Item No. A002. The vendor shall provide, and obtain approval for, the procedures identified below prior to their use. If the vendor considers a required submittal inapplicable to this procurement, the vendor shall submit a written statement justifying the exclusion of the required submittal.

C.3.2.1. ULTRASONIC TEST PROCEDURE

- a. Procedure (per MIL-T-16420 and NAVSEA T9074-AS-GIB-010/271 – for both shearwave and thickness), including acceptance criteria (per MIL-T-16420 and NAVSEA T9074-AS-GIB-010/271), shall be submitted for shearwave and thickness. Ultrasonic calibration standard certification shall also be submitted for review and approval (this may be submitted no later than two weeks prior to scheduling the ultrasonic inspection witness).

C.4. CLEANLINESS REQUIREMENTS:

C.4.1. Cleanliness requirements shall be per Section 3.2 of the IRPOD and paragraph 3.1.3 of MIL-T-16420K with Amendment 1.

C.5. MATERIAL MARKING (LEVEL 1 MATERIAL):

C.5.1. Each piece shall be permanently marked at least once with the information of (a), (b), (c), and (d) below using the permanent marking methods of MIL-STD-792 or the permanent marking method specified in the material specification. If any marking specified below cannot be applied due to space limitation, a waiver request shall be submitted to the contracting officer.

- (a) The Material Designator/Code "CNA".
- (b) The manufacturer's name, trademark, or symbol.

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- (c) The vendor traceability number traceable to all manufacture test and inspection information and to a unique set of physical, mechanical, and chemical test results which represent the material used to make the piece.
- (d) Any information required by the specification or drawing (if applicable) in addition to that required by (a), (b), and (c) above shall be marked either per the specification/drawing or at least once per the permanent marking methods authorized above. Any Material Designator/Code listed in the specification or drawing (if applicable) need not be marked on material, if it is different from that specified by (a) above.

C.5.2. If feasible, consolidate the placement of the permanent marking(s) to accommodate the marking of a nine digit final acceptance marking at Portsmouth Naval Shipyard. Do not compromise the legibility of the permanent markings to meet this request.

C.5.3. The location of the markings shall be on the outer diameter on the end(s) of each piece. Die stamp markings shall not be used for permanently marking pipe or tube per MIL-STD-792.

C.6. NAVSEA HQ C-223-H003 Exclusion of Mercury (NAVSEA) (MAR 2019).

SECTION D - PRESERVATION/PACKAGING/PACKING:

D.1. Marking requirements for packaging shall be per Section 5.0 of the IRPOD.

D.2. Preservation, Packaging, and Packing shall be in accordance with ASTM D3951 and MIL-T-16420 paragraph 5.1 (Level C), as modified below:

- (a) The material shall be shipped in wooden boxes. For 6-inch OD and larger pipe/tube, the wooden boxes shall be constructed of 2-inch thick by 6-inch wide minimum planks nailed or screwed together. The exterior of the boxes shall be banded with metal straps.

D.3. The use of MIL-STD-2073-1 packaging codes are not required for this contract.

SECTION E - INSPECTION ACCEPTANCE:

E.1. Government inspection is required during manufacture and prior to shipment from this facility. Upon receipt of this order, promptly notify the Representative of the Government office who normally serves your facility so that appropriate planning for Government inspection can be accomplished. In the event a Representative of the Government office cannot be located, the purchasing agent should be notified promptly. Provide the location where Procurement Quality Assurance (PQA) of the supplies will be made along with the organization that will be performing PQA.

E.2. All non-conformances to contract requirements affecting product and/or documentation shall require procuring activity waiver approval to accept.

E.2.1. Any request for waivers/deviations should provide written justification for the requested change including an evaluation which demonstrates that the proposed non-conformance will not affect the quality, form, fit, or function of the part.

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E.3. FAR 52.246-11, Higher-Level Contract Quality Requirements

Supplemented by:

NAVSEA HQ H-246-H001 Calibration System Requirements (NAVSEA)

NAVSEA HQ E-246-H020 Quality Management System Requirements (NAVSEA)

NAVSEA HQ C-246-H002 Government use of Contractor's Inspection Equipment (NAVSEA)

(MIL-I-45208 or ISO 9001/9002 with STR-ISO 9000 Supplement A or B or C)

E.4. FAR 52.246-2 Inspection of Supplies-Fixed-Price (Aug 1996).

E.5. FAR 52.213-4 Terms and Conditions – Simplified Acquisition (Other than Commercial Items)